

$$\langle v_a v_b \rangle = (\langle v_a + v'_a \rangle)(\langle v_b + v'_b \rangle) = v_a v_b \quad "v_a \text{ mean } v \text{ of } a"$$

Momentum transport ligning (3)

$$\star \left(\frac{\partial v}{\partial t} + (v \cdot \nabla) v \right) m n = - \frac{\partial p}{\partial x_a} - \frac{\partial \pi_{a\beta}}{\partial x_\beta} + e n (E_a + \frac{1}{c} [vB]_a) + R_a$$

(1) $p = n m \langle v'^2 \rangle / 3 = n T$ "scalar tryk for partikel a"

(2) $\pi_{a\beta} = n m \langle v'_a v'_\beta - (v'^2/3) \delta_{a\beta} \rangle$ "Stress tensor, af røbe for kugle fordeling"

(3) $R = \int m v' \cdot \epsilon \, dv$ "Gennemsnitlige ændring i momentum for kollision mellem a b"

inerti = trykgradient + Viskøs spending + Lorentz kraft + kollisionskraft

Eksempel på hvordan ligningen kan se ud

Symbol type:

konstanter: m_a, e_a, c

Vektorer: V_a, E, B, R, n, p : (x, y, z, t)

Matricer: $\pi_a(x, t) = \begin{pmatrix} \pi_{xx} & \pi_{xy} & \pi_{xz} \\ \pi_{yx} & \pi_{yy} & \pi_{yz} \\ \pi_{zx} & \pi_{zy} & \pi_{zz} \end{pmatrix}$

Varme transport ligning

$$\star \frac{3}{2} \frac{\partial n T}{\partial t} + \text{div} \left(\frac{3}{2} n T V \right) + n T \text{div} V + \pi_{a\beta} \frac{\partial v_a}{\partial x_\beta} + \text{div} q = Q$$

$q = \int \frac{m}{2} v'^2 v f(t, r, v) dv = n m \langle \frac{v'^2}{2} v' \rangle$ "Flux densitet overført med partikler"

$Q = \int \frac{m v'^2}{2} \cdot \epsilon \, dv$ "Varme genereret som konsekvens af kollision"

$$\star \frac{\partial n_a}{\partial t} + \text{div}(n_a v_a) = 0$$

$$\star \left(\frac{\partial v_a}{\partial t} + (v_a \cdot \nabla) v_a \right) m_a n_a = - \frac{\partial p_a}{\partial x_a} - \frac{\partial \pi_{a\alpha\beta}}{\partial x_\beta} + e n_a (E_a + \frac{1}{c} [v_a B]_\alpha) + R_a$$

$$\star \frac{3}{2} n_a \left(\frac{\partial T_a}{\partial t} + (v_a \cdot \nabla) T_a \right) + p_a \text{div} v_a = - \text{div} q_a - \pi_{a\alpha\beta} \frac{\partial v_{a\alpha}}{\partial x_\beta} + Q_a$$